characteristic of real-world happenings. First, the things we encounter in day-to-day life are larger and more complicated than a single photon. Second, the things we encounter in day-to-day life are not isolated: they interact with us and with the environment. The book now in your hands is subject to human contact and, more generally, is continually struck by photons and air molecules. Moreover, since the book itself is made of many molecules and atoms, these constantly jittering constituents are continually bouncing off each other as well. The same is true for pointers on measuring devices, for cats, for human brains, and for just about everything you encounter in daily life. On astrophysical scales, the earth, the moon, asteroids, and the other planets are continually bombarded by photons from the sun. Even a grain of dust floating in the darkness of outer space is subject to continual hits from low-energy microwave photons that have been streaming through space since a short time after the big bang. And so, to understand what quantum mechanics says about real-world happenings—as opposed to pristine laboratory experiments—we should apply Schrödinger's equation to these more complex, messier situations.

In essence, this is what Zeh emphasized, and his work, together with that of many others who have followed, has revealed something quite wonderful. Although photons and air molecules are too small to have any significant effect on the motion of a big object like this book or a cat, they are able to do something else. They continually "nudge" the big object's wavefunction, or, in physics-speak, they disturb its *coherence:* they blur its orderly sequence of crest followed by trough followed by crest. This is critical, because a wavefunction's orderliness is central to generating interference effects (see Figure 4.2). And so, much as adding tagging devices to the double-slit experiment blurs the resulting wavefunction and thereby washes out interference effects, the constant bombardment of objects by constituents of their environment also washes out the possibility of intereference phenomena. In turn, once quantum interference is no longer possible, the probabilities inherent to quantum mechanics are, for all practical purposes, just like the probabilities inherent to coin tosses and roulette wheels. Once environmental decoherence blurs a wavefunction, the exotic nature of quantum probabilities melts into the more familiar probabilities of day-to-day life.[15] This suggests a resolution of the quantum measurement